

- Description

This module manages a smart greenhouse humidity control system, ensuring that:

1. The system continuously monitors the current humidity level (in %).
2. If the humidity drops below the minimum threshold, the system activates the humidifier to increase moisture.
3. If the humidity exceeds the maximum threshold, the system activates the dehumidifier to reduce excess moisture.
4. If the humidity is within the optimal range, no action is taken.

This system helps optimize plant growth conditions by maintaining an ideal humidity level in the greenhouse.

- Specification

module SmartGreenhouseHumidityControl

var currentHumidity, minHumidity, maxHumidity: real

var humidifierActive, dehumidifierActive: boolean

process updateHumidity(newHumidity: real) returns (currentHumidity: real, humidifierActive: boolean, dehumidifierActive: boolean)

functional scenario form:

currentHumidity < minHumidity && humidifierActive = true, dehumidifierActive = false, Humidifier activated.

||

currentHumidity > maxHumidity && humidifierActive = false, dehumidifierActive = true, Dehumidifier activated.

||

currentHumidity >= minHumidity && humidifierActive = false, dehumidifierActive = false, No device activated.

end process

end module

- Java Code

```
package code;
public class c80_SmartGreenhouse {
    private double minHumidity;
    private double maxHumidity;
    private double currentHumidity;
    private boolean humidifierActive;
    private boolean dehumidifierActive;
    public c80_SmartGreenhouse(double minHumidity, double maxHumidity,
double initialHumidity) {
        this.minHumidity = minHumidity;
        this.maxHumidity = maxHumidity;
        this.currentHumidity = initialHumidity;
        this.humidifierActive = false;
    }
}
```

```

        this.dehumidifierActive = false;
    }
    public void updateHumidity(double newHumidity) {
        currentHumidity = newHumidity;
        adjustHumidityControl();
        System.out.printf("Current Humidity: %.2f%%, Humidifier: %s, Dehumidifier: %s\n",
            currentHumidity,
            humidifierActive ? "ON" : "OFF",
            dehumidifierActive ? "ON" : "OFF");
        if (humidifierActive) {
            System.out.println("ALERT: Low humidity detected. Humidifier activated.");
        } else if (dehumidifierActive) {
            System.out.println("ALERT: High humidity detected. Dehumidifier activated.");
        } else {
            System.out.println("Humidity is optimal. No action needed.");
        }
    }
    private void adjustHumidityControl() {
        if (currentHumidity < minHumidity) {
            humidifierActive = true;
            dehumidifierActive = false;
        } else if (currentHumidity > maxHumidity) {
            humidifierActive = false;
            dehumidifierActive = true;
        } else {
            humidifierActive = false;
            dehumidifierActive = false;
        }
    }
    public double getCurrentHumidity() {
        return currentHumidity;
    }
    public static void main(String[] args) {
        c80_SmartGreenhouse greenhouse = new c80_SmartGreenhouse(40.0, 70.0, 50.0);
        // testcase-VT:
        //     greenhouse.updateHumidity(35.0);
        //     greenhouse.updateHumidity(75.0);
        //     greenhouse.updateHumidity(60.0);
    }
}

```

```

        // testcase-FT:
//      greenhouse.updateHumidity(-1.6284984868752742E308);
//      greenhouse.updateHumidity(-1.7976931348623157E308);
//      greenhouse.updateHumidity(2.3264264098218203E307);

        // testcase-Z3:
//      greenhouse.updateHumidity(35.0);
//      greenhouse.updateHumidity(75.0);
//      greenhouse.updateHumidity(60.0);

        // testcase-UVT:
        greenhouse.updateHumidity(35.0);
        greenhouse.updateHumidity(75.0);
        greenhouse.updateHumidity(60.0);
    }
}

```

- Coverage report

1. Test case number: 3
2. V-Method testing branch coverage: 100%

Element	Coverage	Covered Branches	Missed Branches	Total Branches
▼ VibrationTestingTool	1.4 %	12	830	842
▼ src	1.4 %	12	830	842
▼ code	1.4 %	12	830	842
▼ c80_SmartGreenhouse.java	100.0 %	12	0	12
▼ c80_SmartGreenhouse	100.0 %	12	0	12
main(String[])		0	0	0
c80_SmartGreenhouse(double, c		0	0	0
adjustHumidityControl()	100.0 %	4	0	4
updateHumidity(double)	100.0 %	8	0	8

3. Vibration testing instruction coverage: 97.6%

Element	Coverage	Covered Instructions	Missed Instructions	Total Instructions
▼ VibrationTestingTool	0.6 %	121	20,997	21,118
▼ src	0.6 %	121	20,997	21,118
▼ code	0.6 %	121	20,997	21,118
▼ c80_SmartGreenhouse.java	97.6 %	121	3	124
▼ c80_SmartGreenhouse	97.6 %	121	3	124
main(String[])	100.0 %	17	0	17
c80_SmartGreenhouse(double, c	100.0 %	18	0	18
adjustHumidityControl()	100.0 %	33	0	33
updateHumidity(double)	100.0 %	53	0	53

4. Fuzz testing branch coverage: 83.3%

Element	Coverage	Covered Branches	Missed Branches	Total Branches
▼ VibrationTestingTool	1.2 %	10	832	842
▼ src	1.2 %	10	832	842
▼ code	1.2 %	10	832	842
▼ c80_SmartGreenhouse.java	83.3 %	10	2	12
▼ c80_SmartGreenhouse	83.3 %	10	2	12
main(String[])		0	0	0
c80_SmartGreenhouse(double, c		0	0	0
adjustHumidityControl()	75.0 %	3	1	4
updateHumidity(double)	87.5 %	7	1	8

5. Fuzz testing instruction coverage: 90.3%

Element	Coverage	Covered Instructions	Missed Instructions	Total Instructions
▼ VibrationTestingTool	0.5 %	112	21,006	21,118
▼ src	0.5 %	112	21,006	21,118
▼ code	0.5 %	112	21,006	21,118
▼ c80_SmartGreenhouse.java	90.3 %	112	12	124
▼ c80_SmartGreenhouse	90.3 %	112	12	124
main(String[])	100.0 %	17	0	17
c80_SmartGreenhouse(double, c	100.0 %	18	0	18
adjustHumidityControl()	81.8 %	27	6	33
updateHumidity(double)	94.3 %	50	3	53

6. Z3 branch coverage: 100%

Element	Coverage	Covered Branches	Missed Branches	Total Branches
▼ VibrationTestingTool	1.4 %	12	830	842
▼ src	1.4 %	12	830	842
▼ code	1.4 %	12	830	842
▼ c80_SmartGreenhouse.java	100.0 %	12	0	12
▼ c80_SmartGreenhouse	100.0 %	12	0	12
main(String[])		0	0	0
c80_SmartGreenhouse(double, c		0	0	0
adjustHumidityControl()	100.0 %	4	0	4
updateHumidity(double)	100.0 %	8	0	8

7. Z3 instruction coverage: 97.6%

Element	Coverage	Covered Instructions	Missed Instructions	Total Instructions
▼ VibrationTestingTool	0.5 %	121	22,908	23,029
▼ src	0.5 %	121	22,908	23,029
▼ code	0.5 %	121	22,908	23,029
▼ c80_SmartGreenhouse.java	97.6 %	121	3	124
▼ c80_SmartGreenhouse	97.6 %	121	3	124
main(String[])	100.0 %	17	0	17
c80_SmartGreenhouse(double, dout	100.0 %	18	0	18
adjustHumidityControl()	100.0 %	33	0	33
updateHumidity(double)	100.0 %	53	0	53

8. Unimproved V-method branch coverage: 100%

Element	Coverage	Covered Branches	Missed Branches	Total Branches
▼ VibrationTestingTool	1.4 %	12	830	842
▼ src	1.4 %	12	830	842
▼ code	1.4 %	12	830	842
▼ c80_SmartGreenhouse.java	100.0 %	12	0	12
▼ c80_SmartGreenhouse	100.0 %	12	0	12
main(String[])		0	0	0
c80_SmartGreenhouse(double, c		0	0	0
adjustHumidityControl()	100.0 %	4	0	4
updateHumidity(double)	100.0 %	8	0	8

9. Unimproved V-method instruction coverage: 97.6%

Element	Coverage	Covered Instructions	Missed Instructions	Total Instructions
▼ VibrationTestingTool	0.5 %	121	22,908	23,029
▼ src	0.5 %	121	22,908	23,029
▼ code	0.5 %	121	22,908	23,029
▼ c80_SmartGreenhouse.java	97.6 %	121	3	124
▼ c80_SmartGreenhouse	97.6 %	121	3	124
main(String[])	100.0 %	17	0	17
c80_SmartGreenhouse(double, dout	100.0 %	18	0	18
adjustHumidityControl()	100.0 %	33	0	33
updateHumidity(double)	100.0 %	53	0	53